String Master

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes



Rousong the Fox is a string master who can handle a wide variety of strings with ease. For example, he can braid noodles with his tongue. Likewise, he is good with string data types in computer programming.

One night, Rousong has a dream, in which he picks an infinitely long string out of the bowl when eating noodles. On closer inspection, it is a binary string containing only zeros and ones, concatenated by $0, 1, 10, 11, \ldots$. Formally, let's define string $s^0 = 0$, and $s^i = s^{i-1} + (i)_2$ for each integer $i > 0$, where $a + b$ denotes the concatenation of string a and b, and $(i)_2$ denotes the binary form of integer i without leading zeros. Consequently, the infinitely long string Rousong dreamed of is $s^\infty = 011011100101\ldots$.

Since the length of s^∞ is too large, Rousong only wants to focus on the substring from the l-th character to the r-th character, denoted as $s^\infty_{l,r}$. He wants to find the substring of length n within the string $s^\infty_{l,r}$ with the largest lexicographical order. Formally, please find the index i within $l \le i \le r - n + 1$ that maximizes the lexicographical order of $s^\infty_{i,i+n-1}$. Can you help Rousong solve this problem before he wakes up?

For two binary strings of the same length a and b, we say that string a is lexicographically greater than string b if for the first index i where a and b differ, the i-th character of a is 1 and of b is 0.

Input

The first line contains an integer T ($1 \le T \le 100$), indicating the number of test cases.

Each test case contains three integers l, r, n ($1 \le l \le r \le 10^{18}$, $1 \le n \le \min(r - l + 1, 10^6)$) in a single line, indicating the range to be focused on and the length of the substring.

It is guaranteed that $\sum n \le 10^6$ over all test cases.

Output

For each test case, output the substring with the largest lexicographical order in a single line.

Example

standard input	standard output
6 6 13 3 1 9 9 1 1451419198 10 987 6543 21 1123 581321 34 1000010 1000030 18	110 011011100 1111111111 11111111100000000010 11111111111111110000000000000100 000111110011010000

Note

For the first test case of the sample, $s^\infty = 01101\underline{1100101}10111\ldots$, so $s^\infty_{6,13} = 11001011$. The substrings

of length 3 are $110, 100, 001, 010, 101, 011$. The one with the largest lexicographical order is 110.